Probit Regression Function

Applications in Government & Politics

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Topic 1 | Prerequisite Knowledge

Bernoulli Distribution Variance

Used for modeling binary outcomes. The variance is defined as $\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$

General Linearized Model

A *General Linearized Model* is a model that applies a linear function to a dataset, with the general rules of mapping the input to the output y, and mapping it according to a **Link Function**.

Topic 1 | Prerequisite Knowledge

Gradient

A vector of partial derivatives. It is used in mapping the log likelihood function of the linear model. For this case, we use the weights as slopes and apply the transformation $X^T \cdot W$, where W is the weights matrix. This is used in solving for the **Hessian**

Hessian

A matrix of 2nd order partial derivatives. For this case, however, we take the negative Expected Value of the hessian, which simplifies the matrix math. Using the previously defined weights matrix, a formula for solving the Expected Hessian becomes (2):

$$X^T W X^T$$

Topic 1 | Prerequisite Knowledge

Link Function

Topic 2 | Theoretical and Mathematic Explanation

- **Assumptions**:
  - Has all assumptions of a GLM, found in (4)
  - Assumes a Bernouli Distribution for the y data
- **Inputs**:
  - x: the design matrix, defined as having all 1s in the first column vector, with the rest being vectors for the observed values of the data
  - y: the target variable, assumed Bernouli Distribution
  - max_iter: maximum iterations to do
  - tolerance: the level that is closest to zero that decides if there has been a "convergence" to the $\hat{B}$
  - verbose: used for debugging, prints the steps and beta values
  - *note* some of these inputs are handled by glm differently, like the design matrix X is created by just adding a column of 1s to the dataframe passed in as x

Topic 2 | Theoretical and Mathematic Explanation

- **Function**:
  - initialize beta to be a vector of 0s, of length m (number of x columns)
  - for(i in max_iter)
    - get the linear predictor, eta
    - calculate the PDF and CDF of the predictor assuming the normal distribution, defined as $\hat{p}\hat{q}$
    - calculate the ratio of the CDF to the bernouli variance of the PDF $\hat{q}/(\hat{p}(1-\hat{p}))$
    - get a score vector with the previous ratio in the form ratio $\cdot\,(\hat{y}-\hat{p})$
    - calculate the gradient, g, defined by $X^T \cdot$ score
    - now, we calculate the Weight matrix of the form $\frac{\hat{q}}{\hat{p}(1-\hat{p})}$ note the bernouli distribution in the denominator
    - using the Weight vector, we calculate the negative expected Hessian, H, this is written as $-(X^T \cdot W)$
    - finally, a copy of the current beta iteration is made, and the next iteration of beta becomes $\hat{B} = \hat{B} - (H^{-1}g)$ where g is the gradient
    - then the tolerance check is done, checking the squared sum of $\hat{B}_{new} - \hat{B}_{old}$, or $||B_{new} - B_{old}||^2$ if this is less than our tolerance, we have converged

Topic 2 | Theoretical and Mathematic Explanation

- **Usage**:
  - Probit has many uses, but most of them are in econometrics and Government. The closest model in terms of use case is Logistic, as they both follow bernouli variables.
  - The main difference from Logisitc regression is that logistic regression uses the logistic distribution
  - This makes it useful in studying things like human behavior, psychology, economics, and finance.

Arnav Slides Start Here

Topic 3 | Implementation

```r
myprobit = function(X, y, max_iter, tolerance, verbose=FALSE){

  m = ncol(X)
  beta <- rep(0, m)

  for (i in 1:max_iter) {

    eta <- X %*% beta

    phat <- pnorm(eta)
    qhat <- dnorm(eta)

    ratio <- qhat / (phat * (1 - phat))
    score <- (y - phat) * ratio
    g <- t(X) %*% score

    w <- (qhat^2) / (phat * (1 - phat))
    H <- -t(X) %*% (X * as.vector(w))

    copy_beta = beta
    beta <- beta - solve(H, g)

    if (verbose){
      cat("Iteration", i, ":\n")
      j=1
      for (b in beta){
        cat("Beta", j, " = ", beta[j], "\n")
        j=j+1
      }
    }
    if ((sum(beta - copy_beta)^2) < tolerance) {
        return(beta)
        break
      }
  }
}
```

```r
X <- model.matrix(am ~ mpg, data = mtcars)
y <- mtcars$am
myprobit(X,y,32, 0.00009, FALSE)
```

```
##                   [,1]
## (Intercept) -3.9100663
## mpg          0.1822605
```

Solving the System

Optimization Methods

- There are 2 main methods of creating a model for probit regression:
  - 1. Newton-Raphson method We used a method similar to Newton Raphson, but rather than use the Hessian, the Expected Hessian was used, which is used for IRLS and is simpler to calculate.
  - 2. Iteratively Reweighted Least Squares / Fisher Scoring
- These methods can be shown to be mathematically the same(7)

Testing

Topic 4 | Comparison to Other Models

Motivation

Several methods exist for modeling binary outcomes. This section compares probit regression against three alternatives:

1. **Linear Regression (OLS)** — the standard baseline for continuous outcomes
2. **Logistic Regression** — the most widely used GLM for binary data
3. **Lasso Regression** — a penalized method that performs automatic variable selection

The goal is to understand the practical and theoretical differences between these approaches, and to identify when probit regression is the appropriate choice.

Data: Donner Party Survival

We use the **Donner Party survival dataset** from the `Sleuth3` package. The binary outcome is survival status (1 = Survived, 0 = Died), with Age and Sex as predictors.

```r
library(Sleuth3); library(dplyr); library(ggplot2)
library(glmnet);  library(caret)

set.seed(42)

# recode response and sex
donner <- case2001 %>%
  mutate(
    mystatus = if_else(Status == "Survived", 1, 0),
    male     = if_else(Sex == "Male", 1, 0)
  )

head(donner[, c("Age", "Sex", "Status", "mystatus")], 4)
```

Data Preview

```
  Age    Sex   Status mystatus
1  23   Male     Died        0
2  40 Female Survived        1
3  40   Male Survived        1
4  30   Male     Died        0
```

Linear Regression on a Binary Response

OLS fits:

$$\hat{E}(Y \mid \mathbf{X}) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p$$

When $Y \in \{0, 1\}$, this is the **Linear Probability Model (LPM)**. Since $\mathbf{X} \in \mathbb{R}$, fitted values are not constrained to $[0, 1]$, which means the model can produce predicted probabilities below 0 or above 1. This violates the basic requirement for a valid probability model and is the primary motivation for using a link function in GLMs.

OLS on a Binary Response — Code

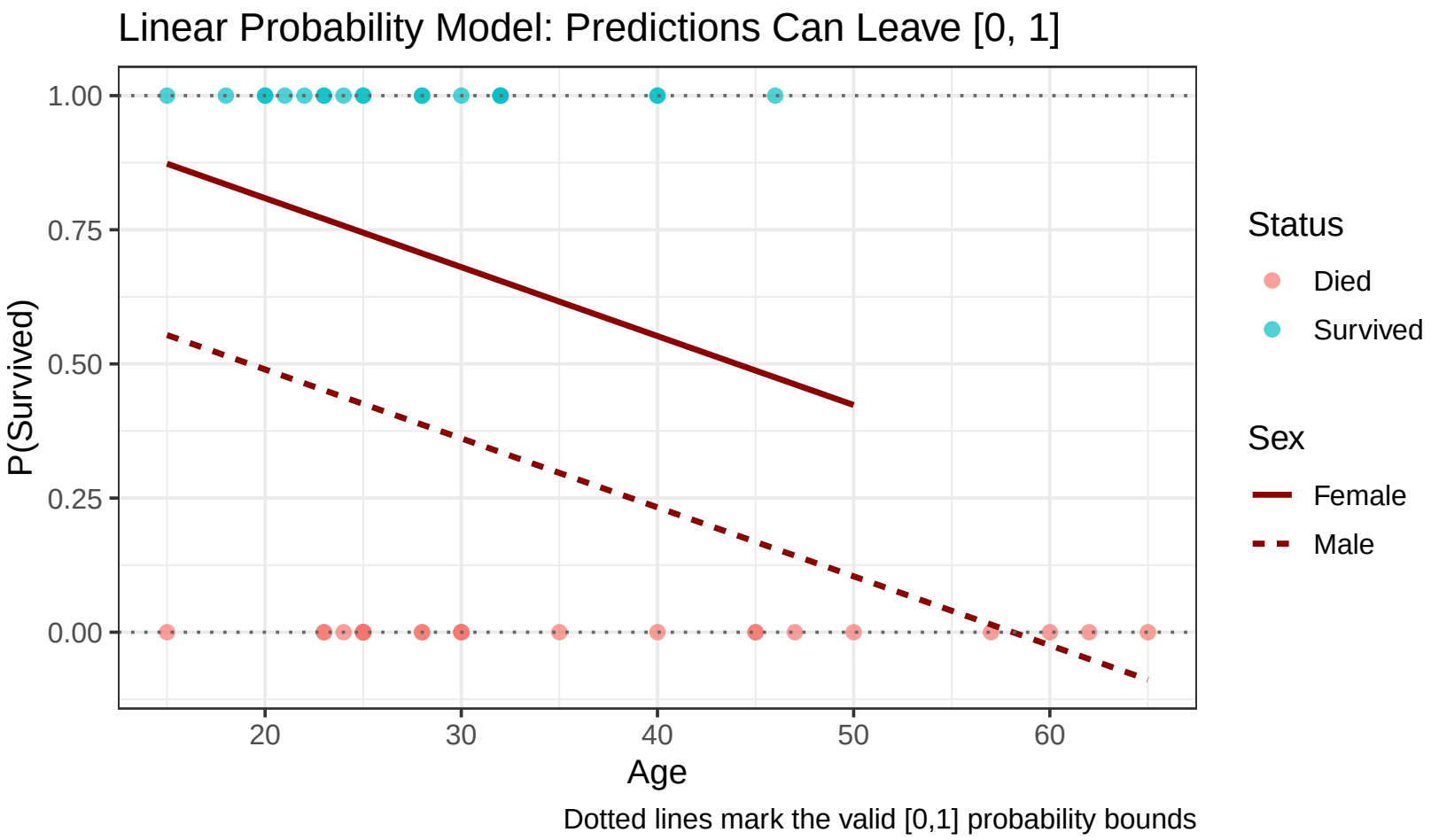
```r
# fit OLS treating the binary outcome as continuous
lpm_fit <- lm(mystatus ~ Age + male, data = donner)

# store fitted values as predicted probabilities (not bounded to [0,1])
donner <- donner %>%
  mutate(pi_lpm = predict(lpm_fit))
```

OLS on a Binary Response — Plot Code

```
ggplot(donner, aes(x = Age, y = mystatus)) +
  geom_point(aes(color = Status), size = 2, alpha = 0.7) +
  geom_line(aes(y = pi_lpm, group = male,
                linetype = factor(male, labels = c("Female","Male"))),
            color = "darkred", linewidth = 1) +
  geom_hline(yintercept = c(0, 1), linetype = "dotted", color = "gray40") +
  labs(title = "Linear Probability Model: Predictions Can Leave [0, 1]",
       y = "P(Survived)", x = "Age", linetype = "Sex",
       caption = "Dotted lines mark the valid [0,1] probability bounds") +
  theme_bw(base_size = 12)
```

OLS on a Binary Response — Plot



OLS Predictions Outside [0, 1]

```r
lpm_preds <- predict(lpm_fit)

cat("Range of OLS predictions:", round(range(lpm_preds), 4), "\n")
```

```
Range of OLS predictions: -0.0885 0.8731
```

```r
cat("Predictions below 0:     ", sum(lpm_preds < 0), "\n")
```

```
Predictions below 0:      3
```

```r
cat("Predictions above 1:     ", sum(lpm_preds > 1), "\n")
```

```
Predictions above 1:      0
```

```r
# misclassification at 0.5 threshold
lpm_class <- ifelse(lpm_preds > 0.5, 1, 0)
cat("OLS misclassification rate:",
    round(mean(lpm_class != donner$mystatus), 4), "\n")
```

```
OLS misclassification rate: 0.2444
```

Even on this small dataset, OLS produces invalid probabilities. Linear regression is not appropriate for binary outcomes.

Logistic Regression

Logistic regression uses the **logit link**:

$$\eta = \text{logit}(\pi) = \ln\left[\frac{\pi}{1-\pi}\right] = \mathbf{x}^T\boldsymbol{\beta} \quad \Rightarrow \quad \hat{\pi} = \frac{e^\eta}{1+e^\eta}$$

This assumes errors follow a logistic distribution.
Probit regression uses the **inverse standard normal CDF**:

$$\eta = \Phi^{-1}(\pi) = \mathbf{x}^T\boldsymbol{\beta} \quad \Rightarrow \quad \hat{\pi} = \Phi(\mathbf{x}^T\hat{\boldsymbol{\beta}})$$

This assumes errors follow a standard normal distribution. Both are valid GLMs for binary outcomes — they differ only in their distributional assumption.

Link Functions — Code

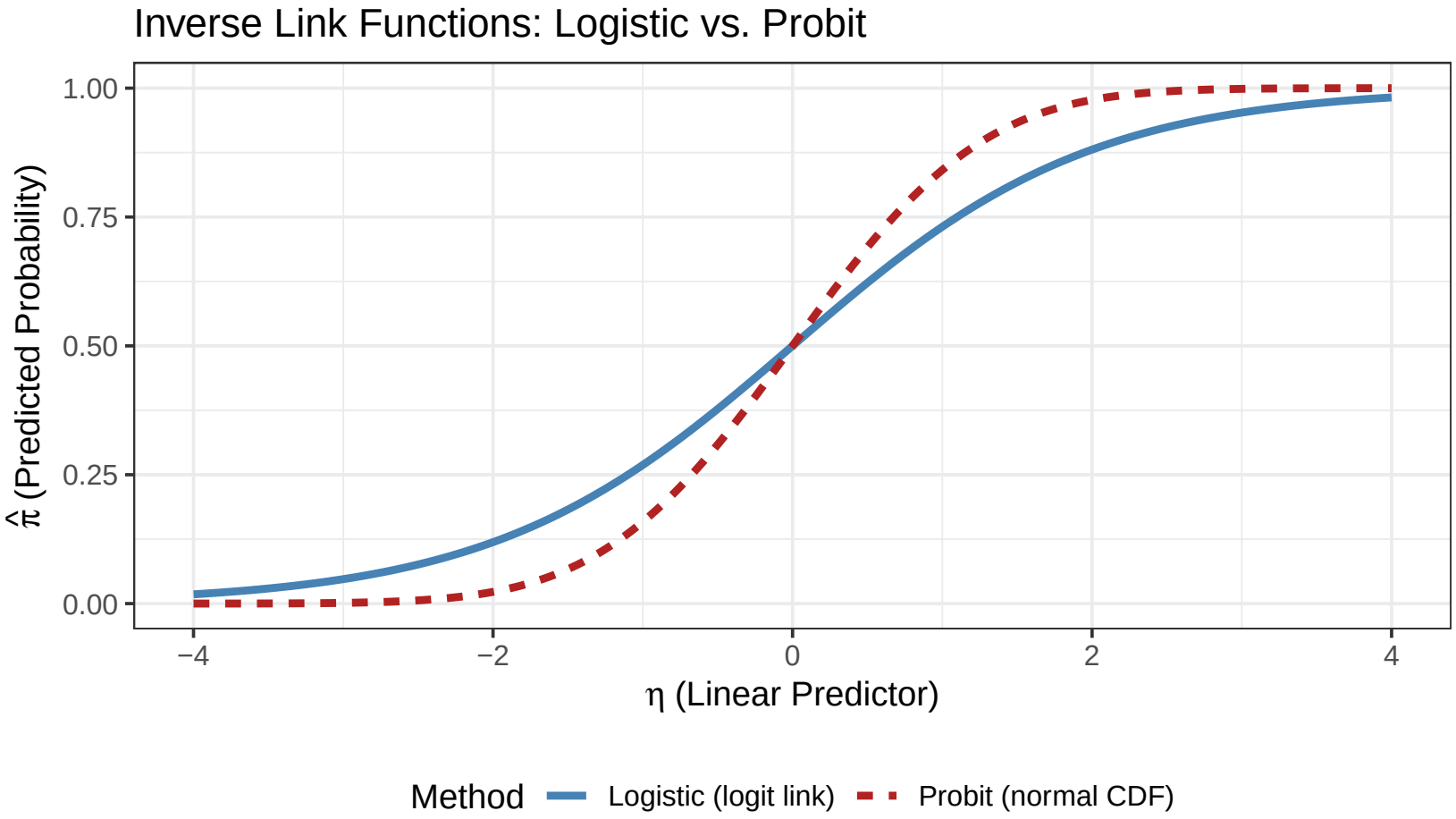
```r
eta_seq <- seq(-4, 4, length.out = 300)

df_links <- data.frame(
  eta    = rep(eta_seq, 2),
  pi     = c(plogis(eta_seq),  # inverse logit
             pnorm(eta_seq)),  # inverse probit (std normal CDF)
  Method = rep(c("Logistic (logit link)", "Probit (normal CDF)"), each = 300)
)
```

Link Functions — Plot Code

```r
ggplot(df_links, aes(x = eta, y = pi, color = Method, linetype = Method)) +
  geom_line(linewidth = 1.3) +
  labs(title = "Inverse Link Functions: Logistic vs. Probit",
       x = expression(eta ~ "(Linear Predictor)"),
       y = expression(hat(pi) ~ "(Predicted Probability)")) +
  scale_color_manual(values = c("steelblue", "firebrick")) +
  theme_bw(base_size = 12) + theme(legend.position = "bottom")
```

Link Functions — Plot



The probit curve has lighter tails — it approaches 0 and 1 faster than the logistic curve.

Fitting Both Models — Code

```r
# fit logistic and probit
logit_fit  <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "logit"))

probit_fit <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "probit"))

# compare coefficients
round(cbind(
  "OLS Coef"    = coef(lpm_fit),
  "Logit Coef"  = coef(logit_fit),
  "Probit Coef" = coef(probit_fit)
), 4)
```

Fitting Both Models — Output

```
            OLS Coef Logit Coef Probit Coef
(Intercept)   1.0658     3.2304      1.9173
Age          -0.0128    -0.0782     -0.0457
male         -0.3192    -1.5973     -0.9583
```

The probit coefficients are smaller by a factor of roughly 1.6 to 1.8. The logistic distribution has variance $\pi^2/3 \approx 3.29$, while the standard normal has variance 1. The scales differ, but the substantive conclusions are the same.

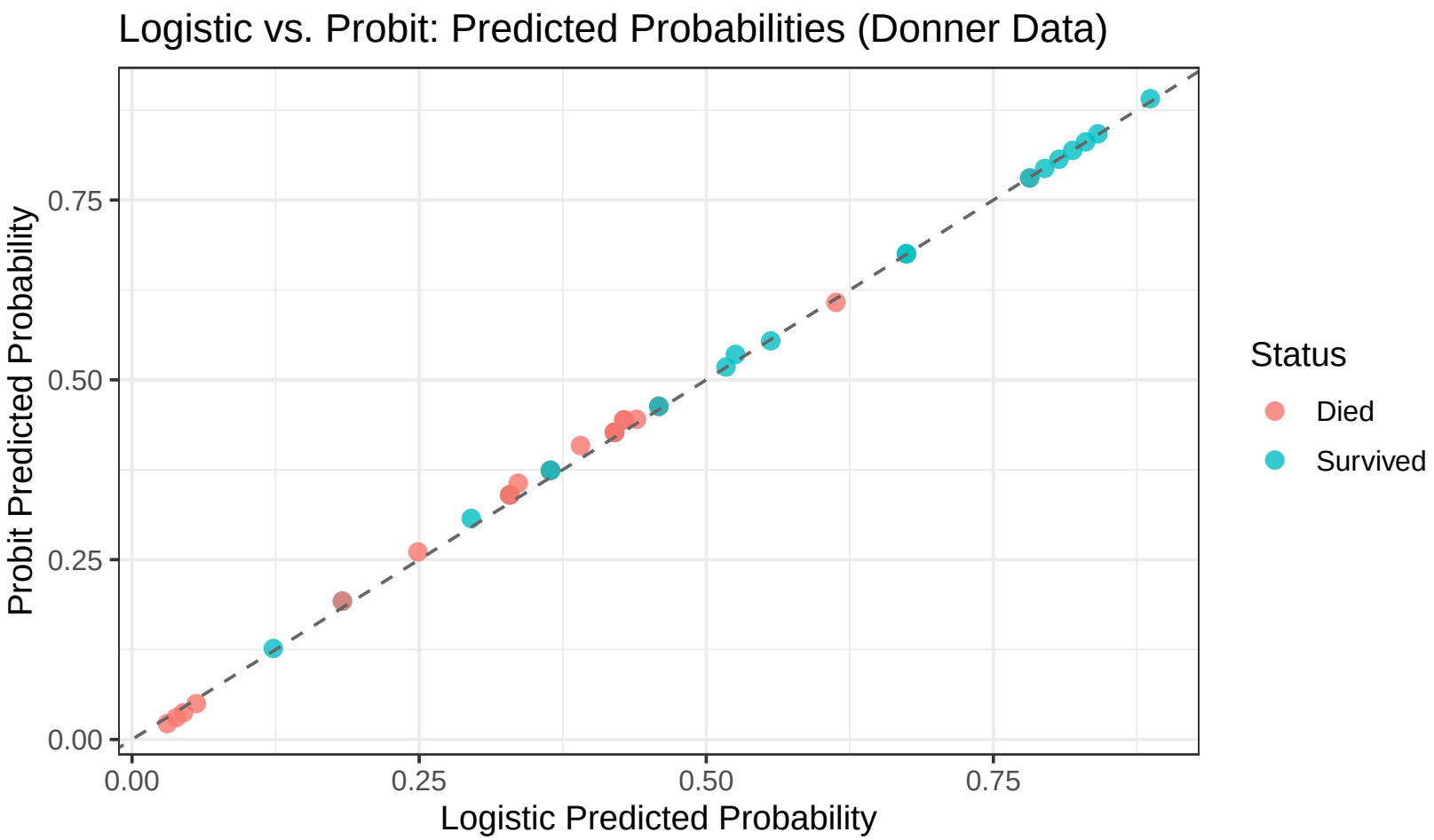
Predicted Probabilities — Code

```r
donner <- donner %>%
  mutate(
    pi_logit  = predict(logit_fit,  type = "response"),
    pi_probit = predict(probit_fit, type = "response")
  )
```

Predicted Probabilities — Plot Code

```r
ggplot(donner, aes(x = pi_logit, y = pi_probit, color = Status)) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray40") +
  labs(title = "Logistic vs. Probit: Predicted Probabilities (Donner Data)",
       x = "Logistic Predicted Probability",
       y = "Probit Predicted Probability") +
  theme_bw(base_size = 12)
```

Predicted Probabilities — Plot



The predicted probabilities from both models are nearly identical, confirming that the choice of link function has minimal practical impact on fitted values.

Model Fit Metrics — Code

```r
cat(sprintf("%-28s %8s %8s\n", "Metric", "Logistic", "Probit"))
cat(sprintf("%-28s %8.3f %8.3f\n", "AIC",
            AIC(logit_fit), AIC(probit_fit)))
cat(sprintf("%-28s %8.3f %8.3f\n", "Residual Deviance",
            deviance(logit_fit), deviance(probit_fit)))
cat(sprintf("%-28s %8.3f %8.3f\n", "Null Deviance",
            logit_fit$null.deviance, probit_fit$null.deviance))

log_err  <- mean((donner$pi_logit  > 0.5) != donner$mystatus)
prob_err <- mean((donner$pi_probit > 0.5) != donner$mystatus)
cat(sprintf("%-28s %8.4f %8.4f\n", "Misclassification Rate",
            log_err, prob_err))
```

Model Fit Metrics — Output

Metric	Logistic	Probit
AIC	57.256	57.283
Residual Deviance	51.256	51.283
Null Deviance	61.827	61.827
Misclassification Rate	0.2222	0.2222

The two models produce nearly identical fit across every metric. AIC differs by less than 0.03, and misclassification rates are equal. The choice between logistic and probit is rarely driven by model performance — it is almost always a question of convention, interpretability, or the underlying distributional assumption.

When to Prefer Probit

1. Latent variable framework. Assume an unobserved variable $Y^* = \mathbf{x}^T\beta + \varepsilon$ where $\varepsilon \sim N(0,1)$. We observe $Y = 1$ if $Y^* > 0$. This is the probit model exactly — a standard framework in economics and political science.

2. Tail behavior. The normal CDF approaches 0 and 1 more rapidly than the logistic distribution. Probit is preferable when probabilities near the extremes are of scientific interest.

3. Field convention. Probit is the standard choice in economics, political science, and related disciplines. For government and political science applications, probit is the conventional model.

4. Marginal effects. Probit coefficients do not have an odds-ratio interpretation, but average marginal effects provide a clear and directly interpretable quantity.

Marginal Effects

Probit coefficients have no direct probability interpretation on their own. The standard approach is to compute **Average Marginal Effects (AME)**:

$$\text{AME}_j = \frac{1}{n} \sum_{i=1}^{n} \phi(\mathbf{x}_i^T \hat{\boldsymbol{\beta}}) \cdot \hat{\beta}_j$$

where $\phi(\cdot)$ is the standard normal PDF.

```r
library(margins)

# compute AMEs
ame <- margins(probit_fit)
summary(ame)[, c("factor", "AME", "SE", "z", "p")]
```

```
 factor     AME     SE       z      p
    Age -0.0148 0.0057 -2.6113 0.0090
   male -0.3109 0.1188 -2.6160 0.0089
```

This gives the change in predicted probability for a one-unit change in each predictor, averaged across all observations.

Lasso Regression

The logistic or probit objective function can be **penalized** to perform variable selection:

$$\hat{\boldsymbol{\beta}} = \text{argmin}_{\boldsymbol{\beta}} \left\{ -\frac{2}{n}\ell(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

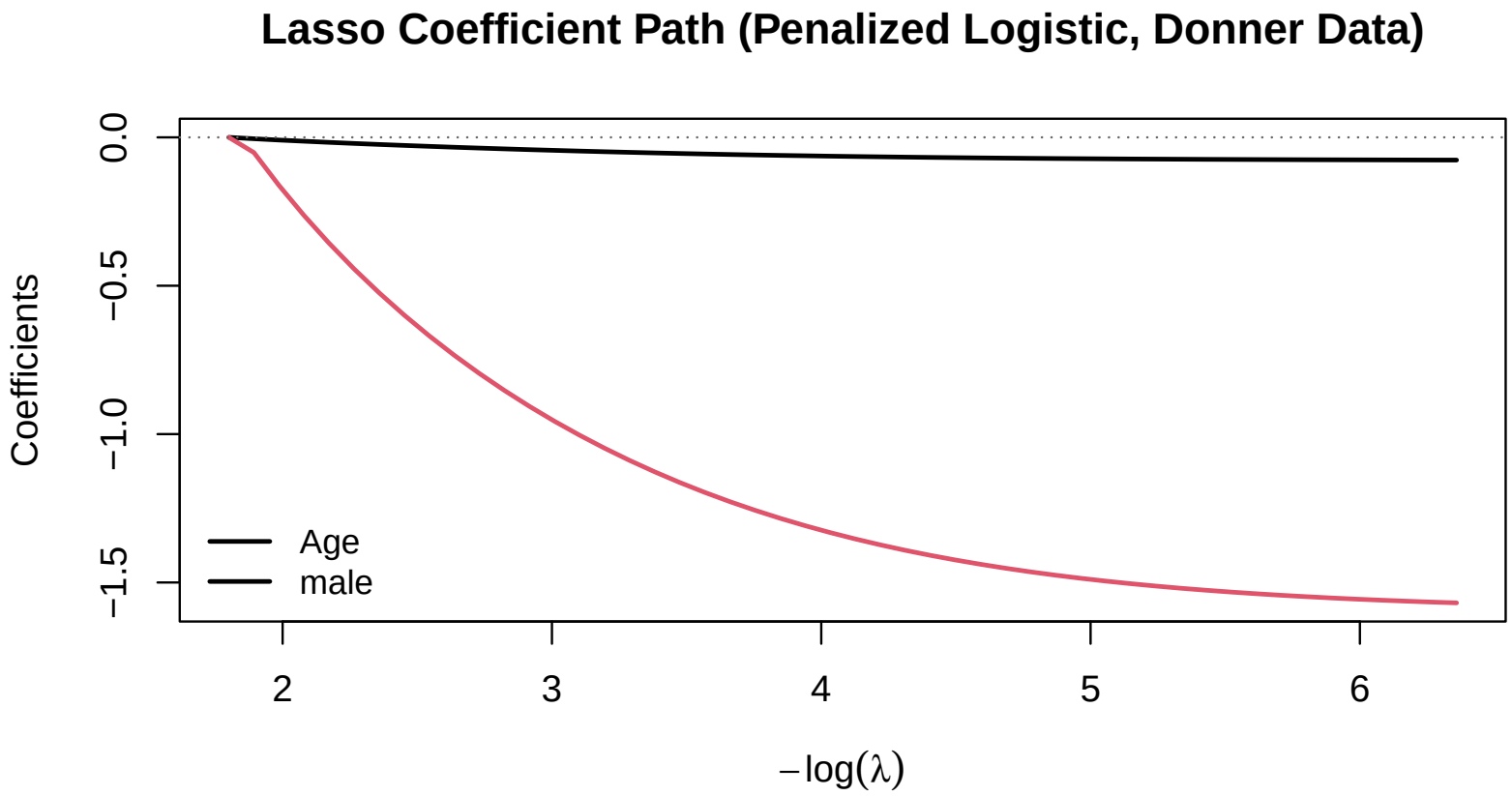
The ℓ_1 penalty shrinks small coefficients toward zero and eliminates irrelevant predictors entirely when λ is large enough. This is particularly useful in high-dimensional settings such as political science data with many demographic and behavioral predictors.

Lasso: Fitting the Model — Code

```r
# glmnet requires a plain matrix, not a model formula
X_mat <- model.matrix(mystatus ~ Age + male, data = donner)[, -1]
y_vec <- donner$mystatus

# fit lasso path over 100 values of lambda (alpha=1 selects lasso over ridge)
lasso_path <- glmnet(X_mat, y_vec, family = "binomial", alpha = 1)
```

Lasso: Coefficient Path



As λ increases (moving left on the x-axis), both coefficients shrink toward zero. The `male` coefficient shrinks more slowly, indicating it is the stronger predictor of survival. `Age` enters the model quickly but with a smaller effect.

Lambda Selection via Cross-Validation — Code

```r
set.seed(42)
# 10-fold CV to select lambda; type.measure="class" minimizes misclassification
cv_lasso <- cv.glmnet(X_mat, y_vec,
                      family       = "binomial",
                      alpha        = 1,
                      nfolds       = 10,
                      type.measure = "class")

# lambda.min: lowest CV error; lambda.1se: most regularized model within 1 SE
cat("Lambda (min CV error): ", round(cv_lasso$lambda.min, 5), "\n")
```

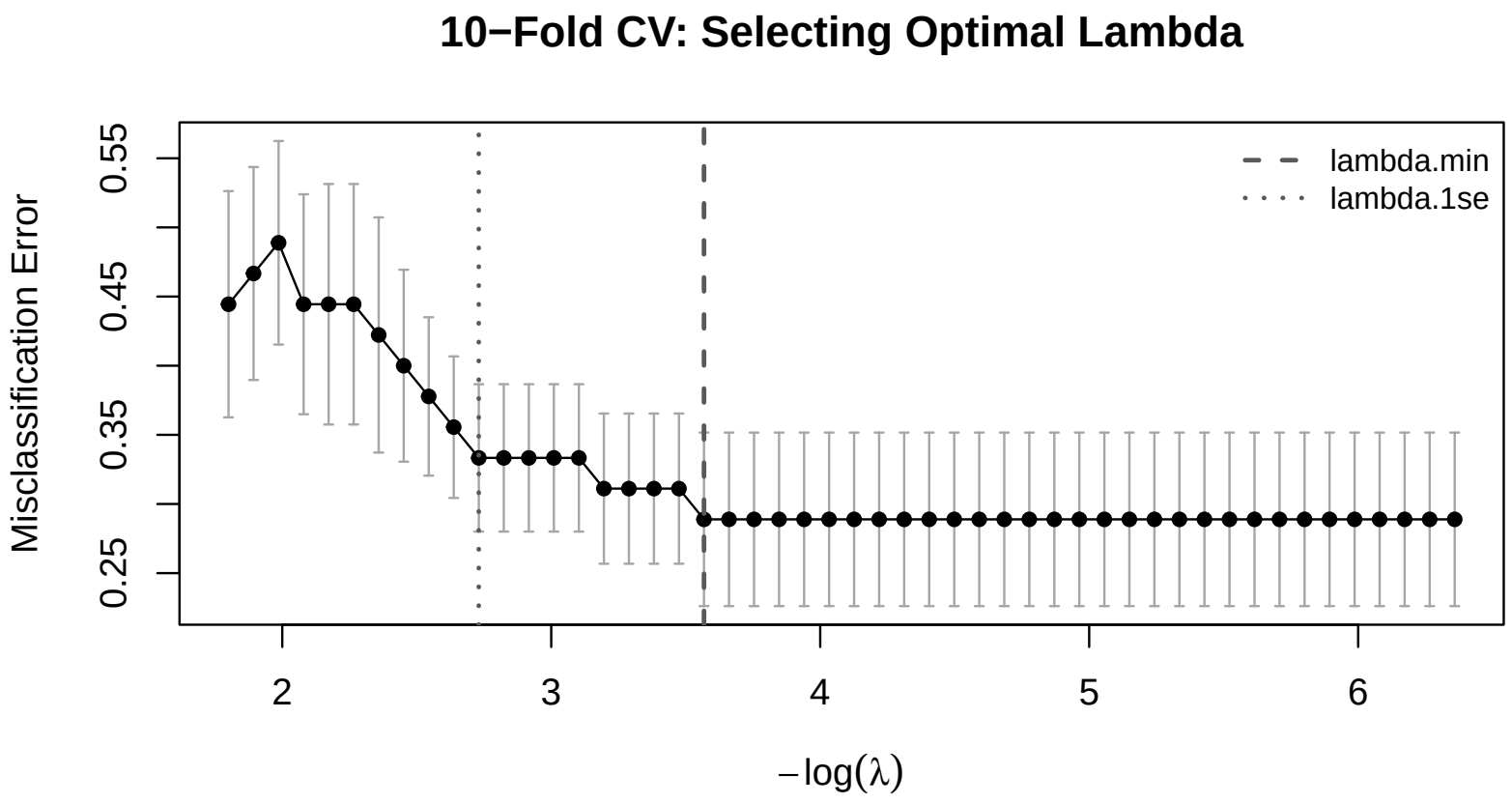
```
Lambda (min CV error):  0.02822
```

```r
cat("Lambda (1 SE rule):    ", round(cv_lasso$lambda.1se, 5), "\n")
```

```
Lambda (1 SE rule):     0.06518
```

Lambda Selection via Cross-Validation — Plot

10-fold CV selects λ_{min} as the value minimizing misclassification error, and λ_{1SE} as the largest λ within one standard error of that minimum — the more parsimonious choice. Error stabilizes well before the full-model limit, indicating that a small penalty is sufficient here.



Lasso Final Coefficients

```r
cat("Lasso coefficients at lambda.min:\n")
```

```
Lasso coefficients at lambda.min:
```

```r
round(as.matrix(coef(cv_lasso, s = "lambda.min")), 5)
```

```
            lambda.min
(Intercept)    2.30441
Age           -0.05615
male          -1.19653
```

```r
cat("\nLasso coefficients at lambda.1se:\n")
```

```

Lasso coefficients at lambda.1se:
```

```r
round(as.matrix(coef(cv_lasso, s = "lambda.1se")), 5)
```

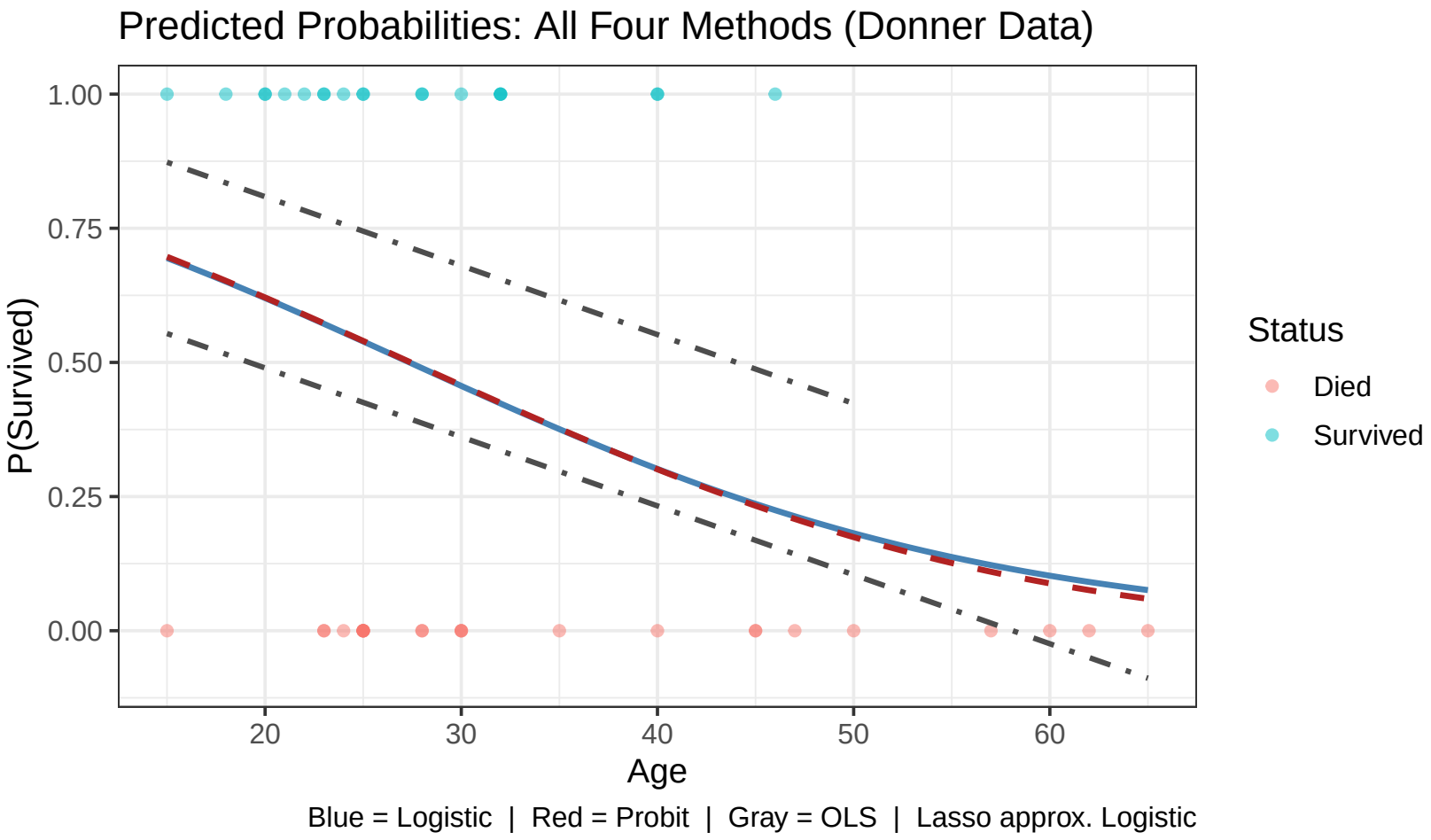
```
            lambda.1se
(Intercept)    1.43295
Age           -0.03615
male          -0.79566
```

A dot (.) indicates a coefficient shrunk exactly to zero — that predictor has been

All Four Methods — Code

```
ggplot(donner, aes(x = Age)) +
  geom_point(aes(y = mystatus, color = Status), alpha = 0.5, size = 1.5) +
  geom_line(aes(y = pi_lpm, group = male), color = "gray30",
            linewidth = 0.9, linetype = "dotdash") +
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "logit")),
              se = FALSE, color = "steelblue", linewidth = 1) +
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "probit")),
              se = FALSE, color = "firebrick", linewidth = 1, linetype = "dashed") +
  labs(title = "Predicted Probabilities: All Four Methods (Donner Data)",
       y = "P(Survived)", x = "Age",
       caption = "Blue = Logistic  |  Red = Probit  |  Gray = OLS  |  Lasso approx. Logistic") +
  theme_bw(base_size = 12)
```

All Four Methods — Plot



Confusion Matrix Comparison — Code

```r
# classify each observation using a 0.5 probability threshold
threshold <- 0.5

results <- donner %>%
  mutate(
    # convert predicted probabilities to binary class labels
    pred_lpm    = factor(ifelse(pi_lpm    > threshold, 1, 0)),
    pred_logit  = factor(ifelse(pi_logit  > threshold, 1, 0)),
    pred_probit = factor(ifelse(pi_probit > threshold, 1, 0)),
    # lasso: get response-scale predictions at the CV-selected lambda
    pred_lasso  = factor(ifelse(
      predict(cv_lasso, newx = X_mat, s = "lambda.min",
              type = "response")[, 1] > threshold, 1, 0)),
    actual = factor(mystatus)
  )
```

Confusion Matrix Comparison — Accuracy Code

```r
# helper: proportion of correctly classified observations
acc <- function(pred, actual) round(mean(pred == actual), 4)

cat(sprintf("%-20s %s\n", "Method",    "Accuracy"))
cat(sprintf("%-20s %s\n", "OLS (LPM)", acc(results$pred_lpm,    results$actual)))
cat(sprintf("%-20s %s\n", "Logistic",  acc(results$pred_logit,  results$actual)))
cat(sprintf("%-20s %s\n", "Probit",    acc(results$pred_probit, results$actual)))
cat(sprintf("%-20s %s\n", "Lasso",     acc(results$pred_lasso,  results$actual)))
```

Confusion Matrix Comparison — Output

```
Method                  Accuracy

OLS (LPM)               0.7556

Logistic                0.7778

Probit                  0.7778

Lasso                   0.7556
```

Logistic and probit tie at 77.78%, while OLS and lasso both fall to 75.56%. The bounded probability models classify slightly better here, though on a dataset of 45 observations the difference is small. Probit and logistic are effectively equivalent in predictive accuracy on this data.

Probit Confusion Matrix

```
          Reference
Prediction  0  1
         0 23  8
         1  2 12

Accuracy             : 0.7778
Sensitivity          : 0.6000
Specificity          : 0.9200
Pos Pred Value       : 0.8571
Neg Pred Value       : 0.7419
Kappa                : NA
```

The probit model is considerably better at identifying deaths (specificity = 0.92) than survivors (sensitivity = 0.60). In the Donner Party data, older males had substantially lower survival odds, and the model captures that pattern well at the majority class. The imbalance between sensitivity and specificity is typical when the outcome is unequally distributed.

Model Comparison Summary

Criterion	OLS (LPM)	Logistic	Probit	Lasso
Predictions in [0, 1]?	No	Yes	Yes	Yes
Link function	Identity	Logit	Φ^{-1} (normal)	Logit or Probit
Error distribution	Normal	Logistic	Normal	Logistic
Estimation method	OLS (closed form)	MLE (IRLS)	MLE (Fisher Scoring)	MLE + ℓ_1 penalty

Model Comparison Summary

Criterion	OLS (LPM)	Logistic	Probit	Lasso
Coefficient interpretation	$\Delta\hat{y}$	Log-odds / Odds ratio	Marginal effects (AME)	Log-odds (shrunk)
Variable selection?	No	No	No	Yes
Tuning required?	No	No	No	Yes (λ)
Preferred when...	Never	General / odds needed	Econ, politics, latent variable	Many predictors
Inference (p-values, CIs)?	Yes	Yes	Yes	Not standard

Key Takeaways

- **Linear regression (OLS)** is not appropriate for binary outcomes. Predicted values are not bounded to $[0, 1]$ and the model is misspecified.
- **Logistic and probit regression** produce nearly identical predicted probabilities and model fit. Coefficients differ in scale by approximately 1.6x due to differing error distributions, but substantive conclusions are the same.
- **Probit is preferred** in economics and political science due to the latent variable framework and field convention. Coefficients are interpreted via average marginal effects.
- **Logistic regression is preferred** when odds ratios are the natural effect measure.
- **Lasso** extends either model with an ℓ_1 penalty for automatic variable selection. Standard p-values and confidence intervals are not available after lasso selection.
- For government and political science applications, probit regression is the standard and appropriate choice.

References

Citations

Comparison Section (Rohan)

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